

Graphing Relations & Linear Forms — Think About Slope!

Mini-Review (read first)

Relation: a set of ordered pairs (x, y) .

Domain: all x -values used. **Range:** all y -values used.

Slope: $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$ (rate of change).

Horizontal line: $y = b$ (slope 0).

Vertical line: $x = a$ (slope undefined; no $y = mx + b$ form).

Slope-intercept form: $y = mx + b$ (b is y -intercept).

Point-slope form: $y - y_1 = m(x - x_1)$.

Two-point to slope: use any known pair(s) to find m , then a form.

x-intercept: solve $0 = mx + b \Rightarrow x = -\frac{b}{m}$ (if $m \neq 0$).

Tip: Choose the form that minimizes algebra; think before you compute.

1. (Domain/Range from pairs) For $R = \{(-2, 3), (0, -1), (4, 3), (4, 7)\}$, give the domain and range.

2. (Restricted line) Consider $y = 2x + 1$ with $x \in \{-2, -1, 0, 1, 2, 3\}$. List the domain and range.

3. (Relation by words) The relation is “all points with $-1 \leq x \leq 4$ and $y \geq -2$.” State the domain and range.

4. (Slope from two points) Find the slope of the line through $(3, -2)$ and $(7, 6)$. Then decide if it rises or falls when x increases.

5. (Rate description $\Rightarrow$ slope) When x increases by 4, y decreases by 10. What is the slope? Is it positive, negative, or zero?

6. (Horizontal/Vertical thinking) Give slope and an equation for each: (a) horizontal line through $(0, -3)$, (b) vertical line through $(4, 5)$.

7. (Build $y = mx + b$ wisely) A line of slope 3 passes through $(2, -5)$. Write its equation in slope-intercept form.

8. (b from one point) In $y = mx + b$, let $m = -\frac{2}{3}$ and the point $(6, 4)$ lies on the line. Find b and the full equation.

9. (Two-point to $y = mx + b$) Find the slope and equation of the line through $(-1, 2)$ and $(5, -4)$ in slope-intercept form.

10. (Point-slope and convert) Through $(-4, 7)$ with slope $-\frac{1}{2}$: write point-slope form, then convert to slope-intercept.

11. (Use point-slope on purpose) Through $(3, -1)$ and $(9, 11)$. First write point-slope form using $(3, -1)$; then give $y = mx + b$.

12. (Graphing relation concept) For the relation $y^2 = x$, is it a function of x ? Give the domain and range over real numbers.

13. (Table: linear or not?)

x	0	1	2	3
y	4	7	10	14

Is the relation linear? Justify via rates of change on each interval and give domain/range.

14. (Slope meaning) A line has y -intercept -3 and slope $\frac{2}{3}$. If x increases by 9, what is the change in y and the new y ?

15. (Same b ; match a point) L_1 : $y = \frac{1}{2}x - 2$. L_2 shares the same y -intercept and passes through $(4, 0)$. Are L_1 and L_2 the same line? Explain.

16. (Verbal $\Rightarrow$ equation) “When $x = 0$, $y = 5$, and for every $+2$ in x , y goes down by 3.”

Write $y = mx + b$. Then solve for x when $y = -7$.

17. (Find m from a form) Suppose $y - 4 = m(x - 1)$ and the line passes through $(-2, 10)$. Find m and then give $y = mx + b$.

18. (Union relation) $R = \{(x, y) \mid y = 2x + 1, -2 \leq x \leq 3\} \cup \{(5, 7)\}$. Give the domain and range.

19. (Do three points line up?) $A(-1, 4)$, $B(2, -2)$, $C(5, -8)$. Are they on one line? If so, give the equation.

20. (Average rate of change) From the table

x	-2	0	3	7
y	1	5	14	26

Is the relation linear? Compute average rate from $x = -2$ to $x = 7$.

21. (x-intercept from m, b) For $y = mx + b$ with $m = -\frac{3}{4}$ and $b = 6$, find the x -intercept and show the computation.

22. (Intercepts given) A line has x -intercept 4 and y -intercept -6 . Find the slope and write $y = mx + b$.

23. (Parametric thinking) Points (a, b) and $(a + 1, b + 3)$ lie on a line. Find the slope and an equation using point-slope; then give $y = mx + b$ in terms of a, b .
